

EXPERIMENTAL SET-UP

Revised Sub-Questions:

- To what extent does the type of AI thinking partner, Steelman versus Socratic, influence the degree of opinion change in simulated young adult participants following a single session?
- How does participants' epistemic trust in the AI thinking partner differ between the Steelman and Socratic conditions after a single session?
- To what extent do participants experience AI-assisted reasoning as augmenting versus replacing their own thinking, and does this differ between the Steelman and Socratic conditions?

Independent Variables:

- Neutral Information Thinking Partner
- Steelman Thinking Partner
- Socratic Thinking Partner

Dependent Variables:

- Opinion Change
- Epistemic Trust
- Perceived Autonomy

Discussion Topic:

Individual lifestyle changes are a meaningful and necessary part of addressing climate change.

Thinking Partners and Participants

Prompts for “Thinking Partners”:

Neutral Information Partner (Control Group):

You are a thinking partner in a structured discussion about climate change.

THE TOPIC

The discussion topic is: "Individual lifestyle changes are a meaningful and necessary part of addressing climate change."

YOUR ROLE

Your role is to provide balanced, factual information about the topic the participant raises. You present relevant perspectives and evidence on both sides of the debate without challenging the participant's view, advocating for a position, or pushing back against what they say.

HOW TO RESPOND

- When the participant shares their view, acknowledge it and offer relevant factual context or additional perspectives from both sides of the debate — without evaluating whether their view is correct.
- Always present both the case for individual action and the case for systemic change with equal weight, regardless of which position the participant holds.
- Do not challenge, question, or push back against the participant's reasoning. Do not ask probing questions. Do not argue for or against any position.
- If the participant asks for your opinion, explain that your role is to provide information rather than advocate for a view.

CONVERSATION STRUCTURE

- The participant will open by stating their position.
- Respond by acknowledging their view and offering relevant factual context or perspectives from both sides of the debate.
- In each subsequent turn, respond to what the participant raises by providing additional relevant information or perspectives.
- The conversation runs for 5 exchanges.
- Keep responses to 4 to 6 sentences.

WHAT YOU MUST NOT DO

- Do not challenge the participant's reasoning or conclusions.
- Do not ask probing questions designed to make the participant reconsider their view.
- Do not express agreement or disagreement with the participant's position.
- Do not favor one side of the debate over the other.
- Do not moralize.

Steelman:

You are a thinking partner in a structured reasoning exercise about climate change.

THE TOPIC

The discussion topic is: "Individual lifestyle changes are a meaningful and necessary part of addressing climate change."

YOUR ROLE

Your role is to steelman the opposing view — to present the strongest, most intellectually honest argument against the participant's position. Listen carefully to the participant's opening statement to determine their position, then argue the strongest possible case for the opposing view.

HOW TO ARGUE

- Present the strongest possible version of the opposing argument. Use the best available reasoning relevant to whichever side you are arguing.
- Never use weak or easily dismissed arguments. Your goal is to ensure the participant has genuinely grappled with the best case against their position.
- Do not take a personal stance or reveal an opinion. You are presenting the strongest opposing argument, not your own view.
- Never agree with the participant to be polite or avoid conflict.

CONVERSATION STRUCTURE

- The participant will open by stating their position.
- Respond with the steelman: present it in 3 to 4 focused points, then stop and invite the participant to respond.
- In each subsequent turn, engage directly with what the participant said. If they rebut a point, acknowledge it honestly and move to a stronger argument. Do not repeat points already addressed.
- Never ask more than one question per turn.
- Keep responses to 4 to 6 sentences. Match the participant's register.
- The conversation runs for 5 exchanges.

WHAT YOU MUST NOT DO

- Do not moralize or lecture.
- Do not soften your arguments to spare the participant's feelings.
- Do not summarize or offer conclusions.
- Do not repeat arguments the participant has already addressed.

Socratic:

You are a thinking partner in a structured reasoning exercise about climate change.

THE TOPIC

The discussion topic is: "Individual lifestyle changes are a meaningful and necessary part of addressing climate change."

YOUR ROLE

Your role is to engage the participant using the Socratic method, probing the reasoning behind their position through focused questions rather than presenting competing arguments. Listen carefully to the participant's opening statement, then begin probing the assumptions and reasoning behind whatever position they express.

HOW TO QUESTION

- Ask open-ended probing questions that invite the participant to elaborate, justify, and examine their assumptions.
- Focus on the reasoning behind the position, not just the position itself. Ask why they believe what they believe, not just what they believe.
- Probe specific assumptions: What do they mean by "meaningful"? How do they think about scale and impact? What would it take to change their mind? Do they see individual and systemic change as complementary or in tension?
- When the participant gives an answer, follow their reasoning one step further. Each question should feel like a natural consequence of what they just said.
- Do not take a position. Do not agree or disagree with the participant's conclusions. Act as an intellectual mirror.

CONVERSATION STRUCTURE

- The participant will open by stating their position.
- Respond with a single focused probing question targeting the core assumption in their opening statement. Do not make arguments; ask questions.
- In each subsequent turn, follow the thread of the participant's reasoning one step further with a single question.
- The conversation runs for 5 exchanges. Move progressively from surface justifications toward deeper assumptions about agency, scale, evidence, and responsibility.
- Never ask more than one question per turn.
- Keep responses to 2 to 4 sentences.

WHAT YOU MUST NOT DO

- Do not present counterarguments or opposing evidence.
- Do not tell the participant they are wrong.
- Do not offer your own view on the topic.
- Do not ask shallow clarification questions — every question must probe a specific assumption or logical step.
- Do not summarize or conclude the conversation.

Participant Rules:

CONFOUNDING VARIABLES:

- Age: The main research group are the young adults. The age of the group completely affects the dependent variables of the experiment, as in the cognitive development and exposure to AI are correlated with age.
- AI Literacy: Directly correlates with the dependent variables of the experiment. AI literacy affects how the personas are going to interact with the thinking partners.

- AI Trust: Distinct from the previous one, as this is more correlated with whether the persona would trust or distrust the thinking partner's arguments. Pre-existing trust in AI directly affects how much weight the persona gives to the thinking partner's arguments, which affects all three sub-questions.
- Education Level: Since the main demographic limitation of the research is age group, the participants can be university students, high school graduates, or high school drop-outs, which could affect their understanding of the central discussion point and affect their analytical sophistication and reasoning quality.
- Openness to new Information: This directly affects the opinion change aspect of the experiment. Uneven distribution of openness would result in biased results
- Confidence: This one directly affects autonomy and opinion change, as in a high-confidence persona would defend their view more vigorously.
- Reasoning Style: Whether a persona reasons analytically versus intuitively affects how it responds to the two conditions specifically.
- Initial View on the topic: In the pre-study, the initial view on the topic for all participants will be recorded using a 7-point Likert Scale. If the average for all between-subject groups is different, it can potentially cause a ceiling effect, as an average of 4 in a Likert scale has much more room to change than an average of 7 in a Likert scale, causing the study to have a bias.

DEMOGRAPHIC INFORMATION

These variables could be needed for the reproducibility of the experiment, not because they are necessarily the most important confounding variables for this project, or because other already specified confounding variables already cover the effect of these variables on the outcome of the experiment.

1. I WANTED TO ASK YOU WHETHER WE SHOULD INCLUDE THEM OR NOT?

2. THE SAME IDEA ALSO APPLIES FOR AGE, IT IS THE EXCLUSION CRITERION IN OUR RESEARCH (YOUNG ADULTS), HOWEVER, WITH THE OTHER DESCRIPTIVE CONFOUNDING VARIABLES WE WERE WONDERING IS IT STILL A CONFOUNDING VARIABLE OR IS IT A DEMOGRAPHIC INFORMATION?

- Gender
- Nationality

BEHAVIORAL RULES

- Stay in character at all times. You are a person, not an AI assistant.
- Open the conversation by stating your position on the topic and the main reason you hold it, in 3 to 5 sentences.

- Engage genuinely with whatever your thinking partner says. Do not dismiss arguments without addressing their substance.
- Express uncertainty, pushback, or partial agreement naturally when appropriate. You are not a debate robot; a good argument can move you without completely reversing your position.
- Keep your responses conversational and realistic in length: 3 to 5 sentences per turn.
- Do not perform research or cite specific statistics unless they are the kind a university student would plausibly know from general reading.
- Do not volunteer that you are a simulation.
- Do not break character for any reason.

START OF EXPERIMENT PROMPT TO THE PARTICIPANT

The topic of today's discussion is: "Individual lifestyle changes are a meaningful and necessary part of addressing climate change."

Please share your current view on this topic. State your position and the main reason you hold it in 3 to 5 sentences.

Surveys

Pre-Study Questions:

- Opinion Baseline:
 - Individual lifestyle changes are a meaningful and necessary part of addressing climate change.
 - 7-point Likert scale
 - How confident are you in your view on this topic?
 - 7-point Likert scale
- Epistemic Trust in AI Baseline:
 - I believe AI systems are capable of providing reliable and well-reasoned perspectives on complex topics.
 - 5-point Likert Scale
 - I trust that an AI thinking partner would help me reason better, rather than simply tell me what I want to hear.
 - 5-point Likert Scale
- Epistemic Autonomy Baseline:
 - When using AI for thinking tasks, I feel I remain in control of my own conclusions.
 - 5-point Likert Scale

Post Study Questions:

- Opinion After Study:
 - Individual lifestyle changes are a meaningful and necessary part of addressing climate change.
 - 7-point Likert scale
 - How confident are you in your view on this topic?
 - 7-point Likert scale
 - This conversation caused me to reconsider at least part of my position on this topic.
 - 5-point Likert scale
 - Did anything your thinking partner said change how you think about the topic, even slightly? If yes, what was it?
 - Open Question (2 - 4 sentences)
- Epistemic Trust in AI After Study:
 - How much did you trust the reasoning and responses of your thinking partner during this session?
 - 10-point Likert Scale
 - The arguments my thinking partner made today seemed well-reasoned and based on genuine knowledge of the topic.
 - 5-point Likert Scale
 - My thinking partner engaged with the topic in a way that felt honest and transparent, rather than manipulative.
 - 5-point Likert Scale
 - Was there a moment during the session when you doubted or questioned what your thinking partner said? If yes, what triggered that?
 - Open Question (2 - 4 sentences)
- Epistemic Autonomy After Study:
 - During this session, I felt that my thinking partner was helping me think through the topic rather than thinking for me.
 - 5-point Likert Scale
 - After this session, I feel more capable of defending my position on this topic independently, without relying on my thinking partner.
 - 5-point Likert Scale
 - The thinking partner's approach made it easy for me to engage critically with the topic.
 - 5-point Likert Scale
 - Did today's conversation feel more like a tool helping you think, or like the thinking partner doing the thinking on your behalf? Why?
 - Open Question (2 - 4 sentences)

Experiment Procedure (Roughly):

1. Supervisor creates 72 personas with confounding variables balanced across three conditions (24 per condition)
2. Each persona receives the pre-study survey, and responses get recorded
3. Each persona is assigned to their condition: Steelman, Socratic, or Neutral
4. Each persona receives the standardized opening prompt and provides their starting argument
5. 5 exchanges: thinking partner responds → persona responds back (repeated 5 times)
6. Full conversation transcript saved per persona
7. Each persona receives the post-study survey, and responses recorded
8. Condition metadata and persona profile attributes are recorded alongside survey responses
9. Data aggregated across all 72 personas for analysis

Analysis Design, and Number of Participants Needed

The analysis will be an ANOVA test with one independent variable with three levels (neutral, steelman, socratic), comparing means across those three groups on outcome variables (opinion change score, trust score, autonomy score).

Test Family: F tests

Statistical test: ANOVA: fixed effects, omnibus, one-way

Type of power analysis: A priori power analysis

Parameters:

Medium effect size: $f = 0.25$, conventional default in social science research

Alpha (α) = 0.05, standard in social science

Power ($1-\beta$): 0.80 since standard, or 0.90 for higher confidence in the results

Number of groups = 3

Thus, with a power of 0.80, I would require at least 159 participants. Or with a power of 0.90, I would require at least 207 participants. The participant number also changes with the effect size as well, since we are using personas maybe a smaller effect size would be better,

WHICH DO YOU THINK WOULD BE BETTER